

Network Infrastructures

Four October lectures explained for a two day study plan

This guide takes you from a basic understanding of networks to the theory covered in NI1–NI4: functional areas, access technologies, fiber deployment, wireless and satellite access, ADSL transmission, VDSL and PPP. It is written in English because the class quizzes are in English.

The thread connecting the lessons

An access network must reach users economically. Reusing copper saves deployment costs, but limits transmission quality. More bandwidth can carry more bits, while attenuation and interference make that bandwidth harder to use. Fiber, DMT and vectoring solve different parts of this problem.

How the material was combined

NI1–NI4 define the scope and lecture sequence. The professor's slides supply terminology, diagrams and course values. The full student summary supplies selected explanations and a useful optical illustration. Repeated topics are merged. Administrative details, unrelated later topics and practice questions are omitted.

The December quiz was inspected only for the style of its questions and answer choices. Its syllabus and marked answers were not used to define this guide. Close distinctions are explained throughout, without reproducing quiz questions.

How to study

Read the explanations before memorising the boxes. When you reach a figure, follow the caption and trace the signal path or axes. Work through the numerical examples with a calculator. Use the final revision pages during your second day.

Technical clarifications are included where the notes use an oversimplified explanation. Values described as historical, illustrative or idealised are not universal performance guarantees.

Your ten hour study plan

Day and time	Reading	Purpose
Day 1 · 30 min	Pages 3–4	Learn units, layers and functional areas
Day 1 · 60 min	Pages 5–7	Understand media, FTTx and optical layouts
Day 1 · 75 min	Pages 8–11	Wireless, mobility, 5G and spectrum
Day 1 · 45 min	Pages 12–13	Satellites and the copper local loop
Day 1 · 90 min	Pages 14–18	Shannon, band plans and interference
Day 2 · 80 min	Pages 19–23	DMT, water filling, QAM and BER
Day 2 · 55 min	Pages 24–26	ADSL devices, unbundling and stack
Day 2 · 65 min	Pages 27–29	VDSL, duplexing and vectoring
Day 2 · 45 min	Pages 30–32	PPP framing, operation and authentication
Day 2 · 55 min	Pages 33–36 and difficult earlier pages	Consolidate definitions and causal chains

The allocations total **600 minutes of actual study**, split into five hours per day. Add short breaks outside this budget. You do not need to learn every acronym with equal depth: first master each device's role and the reason each technology exists.

Read at two levels

First pass: understand the main explanation and the figure. **Second pass:** learn exact distinctions, formula variables and the highlighted course values. The final glossary is a reference, not a substitute for understanding.

Highest priority

Access versus core; FTTx versus AON/PON; wireless versus mobile; latency versus bit rate; frequency versus bandwidth; NEXT versus FEXT; DMT versus QAM; ATU versus DSLAM; FDD versus TDD; LCP versus NCP; PAP versus CHAP.

The vocabulary you need first

A network carries information between devices. Information is represented as **bits**, binary values 0 and 1. A **packet** is a structured block of data with control information, such as addresses. A **frame** carries a packet across a particular link. A **signal** is the physical electrical, optical or radio waveform that carries those bits.

Quantity	Meaning	Unit
Bit rate	Bits transmitted per second	bit/s, kbit/s, Mbit/s
Throughput	Useful data delivered per second	bit/s
Latency	Time until data or a response arrives	ms or s
Frequency f	Oscillations per second	Hz
Bandwidth B	Width of the used frequency interval	Hz
$SNR = S/N$	Signal power divided by noise power	Ratio, or dB after conversion
BER	Fraction or probability of erroneous bits	Dimensionless

A 100 Mbit/s connection can transfer a lot of data per second but can still respond slowly. Conversely, a low latency connection can react quickly without having a large bit rate.

Capacity is an ideal upper limit; an advertised access rate and real application throughput are different quantities.

Units and directions

1 kHz = 1,000 Hz; 1 MHz = 1,000,000 Hz; 1 GHz = 1,000,000,000 Hz. One byte contains eight bits. A 10 MB file contains about 80 Mbit under decimal units. At 8 Mbit/s its ideal transfer time is 10 seconds, before overhead.

Upstream goes from the user to the provider. **Downstream** goes from the provider to the user. These directions describe traffic, not high or low frequency: their frequency allocation depends on the technology.

Layers

Layer 1 deals with physical signals. Layer 2 carries frames over a link. Layer 3, usually IP, handles addressing and packet forwarding across networks. A modem converts between bits and physical signals; a router forwards IP packets; an access point provides Wi-Fi access. A home box often combines all three functions.

A digital service still uses a physical waveform. “Digital data” does not mean the cable carries abstract 0s and 1s without modulation.

LESSON 1

Access metro core and edge

Reading basis: NI1 pp. 3-7; slides 01 pp. 3-6 and 10-11

Imagine sending a file from your home to a friend in another city. Your home connection first reaches the provider through the **access network**. A **metropolitan network** aggregates traffic from a city or urban area. The **core**, also called the backbone or transport network, carries large aggregated flows between areas. The destination is reached through its own metro and access networks.

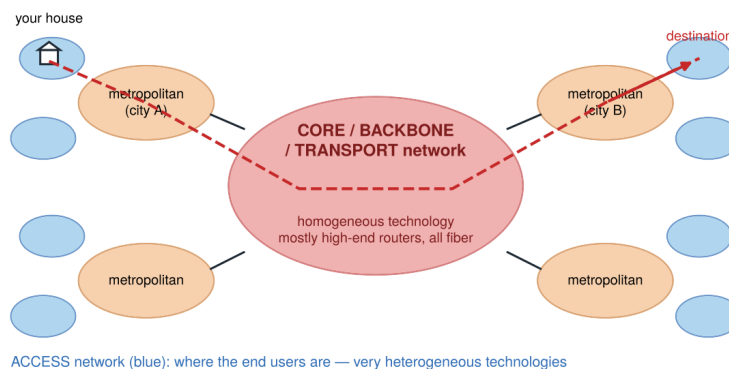


Figure 1. Original diagram from NI1, PDF p. 3. Follow the dashed route from one home, through the orange metro and red core, to the other access area.

The access network connects subscribers to their immediate service provider. The subscriber is the customer using the service. The core interconnects network nodes and areas rather than providing a separate last link to every home. Optical links and high capacity routers make the core relatively homogeneous; access is heterogeneous because homes, offices, moving phones and remote sites have different constraints.

Two uses of the word edge

The slides call the boundary of the core the **network edge**: it may classify traffic and apply policies before traffic enters the backbone. Separately, access plant may be divided into a feeder or distribution part and a final drop or edge part near the subscriber. Interpret “edge” from the diagram and context.

The MPLS example on slide 6 illustrates that some decisions can occur at ingress and packets can then be forwarded through the core using labels. For these four lectures, learn the functional distinction; detailed MPLS operation belongs to later material.

Access reaches users. Metro aggregates an area. Core transports aggregated traffic. The Internet interconnects many providers' networks; it is not one centrally owned backbone.

Why several access technologies coexist

Reading basis: NI1 pp. 4–8; slides 01 pp. 7–18

Providing access is a deployment problem as well as a transmission problem. An operator must reach a large number of geographically scattered users. A fast technology can be unsuitable if its installation cost is too high, its range is too short or it cannot support the required mobility.

Technology or medium	What carries the signal	Why it matters
Optical fiber	Light in a fiber	High capacity; new infrastructure may be needed
DSL	Electrical signals on a telephone twisted pair	Reuses existing copper; rate depends on line quality
Cable access	Electrical signals on coaxial cable	Reuses a different copper infrastructure
Powerline	Electrical wiring carrying data signals	Avoids new home data cables; performance varies
Terrestrial wireless	Radio between ground devices	Avoids a wired final segment; shared radio resources
Satellite	Radio through satellite systems	Reaches remote areas; geometry affects delay

These are overlapping categories, not nine independent alternatives. Fiber and DSL are wired access. Cellular and Wi-Fi are terrestrial wireless technologies. 5G is a cellular generation. Satellite access also uses radio, but its architecture and propagation distances justify treating it separately.

Why the old telephone network matters

The telephone network reached an enormous number of homes and businesses. Its hierarchy of local, regional and long distance networks anticipates access, metro and core. A **Central Office** (CO), or local exchange, is a site where subscriber lines terminate and equipment connects them to the wider network.

Historical slide figures for Telecom Italia illustrate scale: about 33.6 million terminations, 11,000 COs, 150,000 cabinets and over 105.7 million km of copper pairs. These are historical course figures, not a current inventory. DSL's economic motivation is to reuse that already installed copper rather than replace every line immediately.

Do not confuse a telephone twisted pair with a coaxial cable. Both use copper, but their construction and transmission technologies differ.

FTTx tells you where fiber stops

Reading basis: NI1 pp. 8–9; NI4 pp. 9–11; slides 01 pp. 19–26

FTTx means Fiber To The x. The last letter identifies the point reached by the optical segment. It describes deployment reach, not the modulation scheme and not whether the optical network is active or passive. Read each architecture from the CO towards the user.

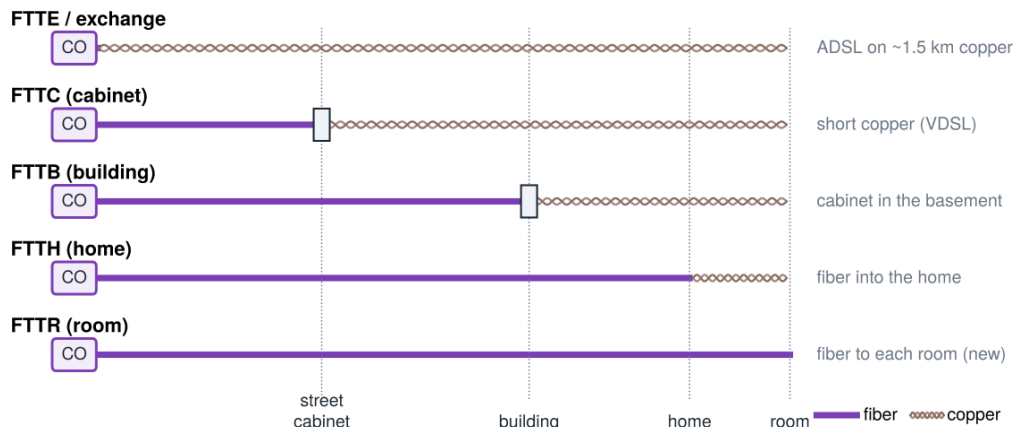


Figure 2. Original NI1 diagram, PDF p. 8. Purple is fiber and brown is copper. Moving down the figure brings fiber progressively closer to the user.

Name	Fiber reaches	What can remain afterwards
FTTE	Exchange or Central Office	Long copper local loop, for example ADSL
FTTC / FTTCab	Street cabinet or curb	Short copper loop, often VDSL
FTTB	Building	In-building copper, Ethernet or another distribution medium
FTTH	Home	Optical access reaches the dwelling
FTTR	Room	Optical distribution extends inside the premises

Bringing fiber closer generally improves the potential rate by reducing the copper length. It does not specify a guaranteed speed: equipment, service provisioning and sharing still matter. FTTH removes telephone copper from the access path to the home, although the home can still use copper Ethernet or Wi-Fi internally.

FTTC can be sold as a fiber service while retaining a copper final segment. FTTH reaches the home with fiber. FTTB reaches the building, which is not necessarily every apartment.

Dedicated fiber active nodes and passive splitters

Reading basis: NI1 pp. 9–12; slides 01 pp. 20–22; full summary pp. 3–4

Fiber reach and fiber distribution are separate choices. The three layouts in the lecture compare dedicated fibers from the exchange with branching through an active node or a passive splitter.

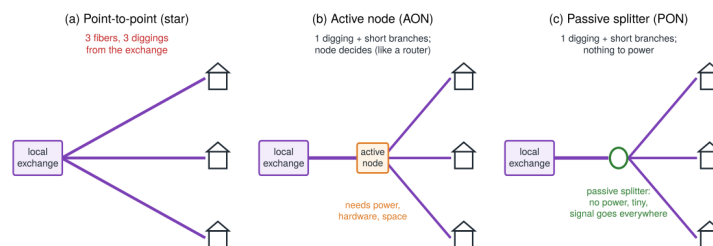


Figure 3. Original NI1 diagram, PDF p. 9. Compare the dedicated branches with a common feeder followed by active or passive branching.

Dedicated point-to-point: each endpoint has its own fiber connection from the exchange.

AON: an active node receives traffic and forwards it to the appropriate branch; it needs electrical power, electronics, maintenance and protected space. **PON:** a passive optical splitter distributes optical power into branches without powered forwarding at that branching point.

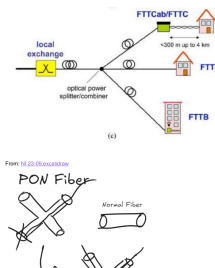


Figure 4. Original optical layout and hand drawing from the full student summary, PDF p. 4. The sketch illustrates passive splitting; it does not represent a powered router.

In the optical diagram, **OLT** is the Optical Line Terminal at the provider side; **ONU** is the Optical Network Unit; **ONT** commonly refers to a subscriber optical termination; **ODN** is the Optical Distribution Network of fibers and passive components. Learn these labels to read the figure, without extending into later PON protocol details.

Civil works such as excavation, ducting and site preparation often dominate fiber installation cost. Sharing a feeder can reduce duplicated infrastructure, but dedicated fibers can also share one trench: a fiber is not inherently a separate excavation. An active branching node adds power, equipment and site costs. A passive splitter avoids powered electronics at that location, but still needs an enclosure and introduces optical power loss. Downstream light is distributed among branches; the splitter does not decide which packet belongs to which user.

“Passive” applies to the distribution segment and splitter. The OLT and ONU/ONT at the ends still need power. A PON splitter neither routes packets nor amplifies the signal.

Fixed wireless access and cellular access

Reading basis: NI2 pp. 2-4; slides 01 pp. 33-34

Fixed Wireless Access (FWA), also called a Wireless Local Loop (WLL), replaces the final wired segment with a radio connection. A provider radio site communicates with equipment at a fixed customer location. The service can avoid the cost of digging a new fiber route. Radio equipment, spectrum and propagation constraints still have costs.

Fixed describes the intended customer connection. A roof terminal connected to an operator site is not the same service model as a smartphone moving between cells. Some FWA implementations use cellular technology, but the user's access service is fixed.

What cellular adds

A cellular network combines **wireless access** with **mobility support**. A cell is the area served by a base station. The hexagons in diagrams are idealised coverage shapes; real coverage depends on terrain, buildings and antennas. Many users share the radio resources of a cell.

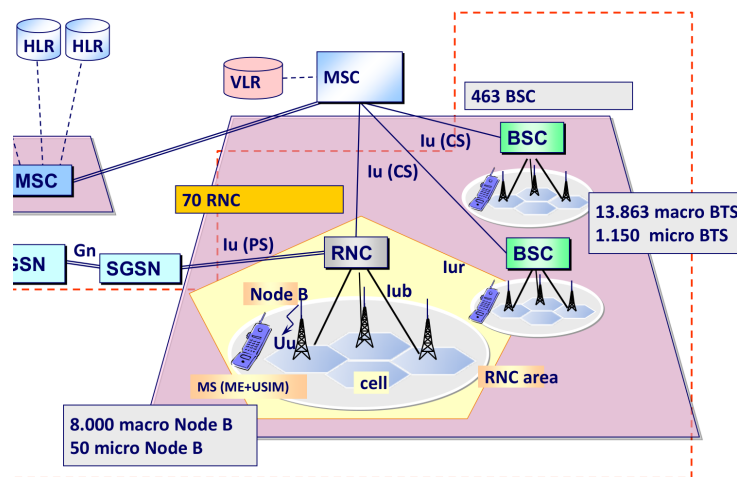


Figure 5. Original professor slide, slides 01 PDF p. 34. Follow Node B to RNC on the 3G side and BTS to BSC on the 2G side. The drawing also contains legacy databases and packet-domain nodes; their detailed operation is outside this block.

The base station is more than an antenna: it is a network element with radio processing and control functions. The simplified lecture architecture uses controllers to coordinate radio access and a mobile switching centre to support legacy switching and mobility.

Wireless does not automatically mean mobile. The UE-to-base-station link is radio; the rest of the simplified course architecture is wired. In real deployments backhaul can also use radio links.

Generations mobility and handover

Reading basis: NI2 pp. 3–6; slides 01 pp. 34–39

Generation	Name to recognise	Main course association
1G	Analog cellular	Analog radio voice
2G	GSM; GPRS and EDGE extensions	Digital cellular; circuit voice plus packet data extensions
3G	UMTS; HSPA extensions	Radio evolution; circuit voice and packet data domains
4G	LTE	Packet based access and core
5G	5G NR and 5G core	Packet services and differentiated requirements

In 2G the base station is a **BTS** (Base Transceiver Station) and its controller a **BSC** (Base Station Controller). In 3G they are **Node B** and **RNC** (Radio Network Controller). The **MSC** is the Mobile Switching Centre in the legacy circuit-switched domain. Mobility is coordinated by several nodes and databases, not by an antenna acting alone.

Nomadic versus mobile

Nomadic use means connecting in one location, moving, then using the network again elsewhere. It does not promise session continuity while moving. Mobile use requires the infrastructure to preserve service as the device changes its serving cell. **Handover** transfers the ongoing connection to a new serving radio point. **Roaming** means using a visited operator's network through an agreement with the home operator. They are different concepts.

Circuit versus packet switching

Circuit switching reserves communication resources for a call. Packet switching sends separate packets using shared resources as traffic arrives. This suits bursty web and data traffic. The lecture describes the evolution from voice-centred 2G/3G to packet-centred 4G/5G; this does not mean 3G had no packet data.

A download, a video stream and an IP voice call all produce packets. A packet network can carry all of them, but their delay and throughput requirements differ. Keeping a session active while moving requires changes to the forwarding path and radio connection.

GSM is 2G; UMTS is 3G; LTE is 4G. Newer devices can support older systems through multiple radio capabilities, but a new generation is not automatically radio-compatible with every older one.

The 5G user plane and control plane

Reading basis: NI2 pp. 6–8; slides 01 PDF p. 35

The simplified 5G diagram separates the path carrying your data from the functions controlling connectivity. **UE** means User Equipment. The **gNB** is the 5G NR base station, part of the radio access network. The **5GC** is the 5G core.

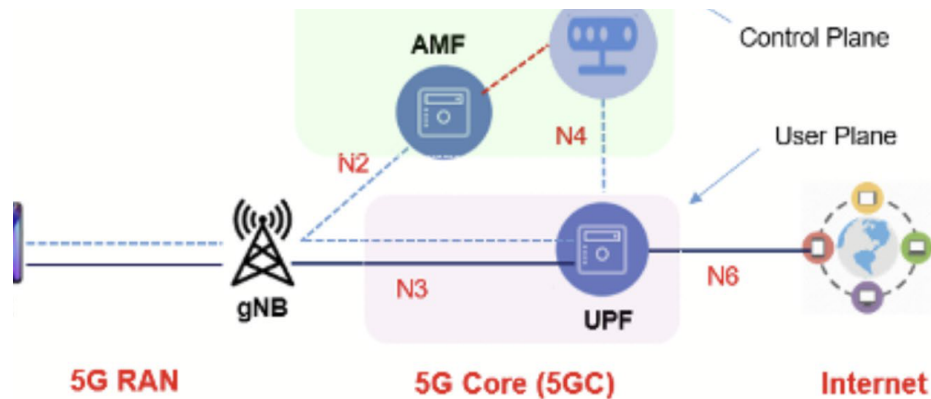


Figure 6. Original 5G diagram from slides 01, PDF p. 35. Trace UE → gNB → UPF → Internet for user traffic; then trace AMF and SMF connections for control.

Function	Remember its job
AMF	Access and Mobility Management Function: registration, connection and mobility management
SMF	Session Management Function: establishes and manages data sessions and controls the UPF
UPF	User Plane Function: forwards and processes user traffic towards the data network

The **user plane** carries user packets. The **control plane** carries signalling that sets up and manages their service. N2 connects access-network control to the AMF; N3 carries user traffic between gNB and UPF; N4 carries SMF control of the UPF; N6 connects the UPF to the external data network.

The SMF must control the user plane because a session's rules need to be enforced on actual packets. Session establishment and packet forwarding are different responsibilities. Likewise, the AMF coordinates access and mobility but is not the radio scheduler inside the gNB.

Precision note

The notes group authentication and encryption under session management. In the full 5G system these involve additional functions and access/security procedures, including AUSF and AMF. Do not learn "SMF encrypts all user data" as a literal definition. The three-function diagram is an introduction, not the entire core.

Service requirements spectrum and Wi-Fi

Reading basis: NI2 pp. 7–12; slides 01 pp. 39–41

The three 5G service families are **eMBB** (enhanced Mobile Broadband), **URLLC** (Ultra-Reliable Low-Latency Communications) and **mMTC** (massive Machine-Type Communications). Their emphases are respectively large data rates, reliable fast responses and support for many devices. A service can have several requirements at once.

Network slicing creates logical networks with different characteristics over shared infrastructure. It is more than putting each application in a separate frequency band: functions and resources can be configured for different service needs.

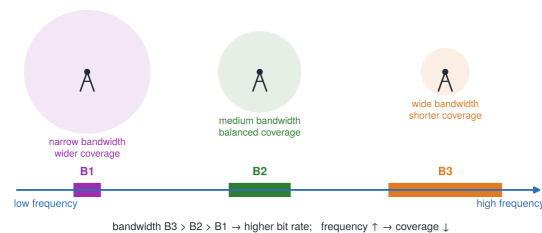


Figure 7. Original NI2 spectrum sketch, PDF p. 9. The horizontal position denotes frequency, interval width denotes bandwidth and the circles indicate qualitative coverage.

Frequency is the location of a band; bandwidth is its width. A 20 MHz band is 20 MHz wide whether centred near 800 MHz or 3.5 GHz. More usable bandwidth can support a larger bit rate at comparable SNR. Lower frequencies generally provide wider coverage under comparable conditions; higher frequencies can offer wider available bands but need denser sites or stronger directional techniques. Frequency alone does not determine bit rate.

Family	Coverage category	Course example
IEEE 802.15	Personal area	Bluetooth family
IEEE 802.11	Local area	Wi-Fi
IEEE 802.16	Metropolitan area	WiMAX
IEEE 802.22	Regional area	WRAN

3GPP develops cellular standards; IEEE 802 includes the wireless families above. Wi-Fi was introduced for local access, without cellular-style wide-area mobility. Modern Wi-Fi can support roaming mechanisms, but continuity depends on deployment and client support.

The course Wi-Fi sequence is 802.11 (1997, 2 Mbit/s), 802.11b (1999, 11 Mbit/s), 802.11g (2003, 54 Mbit/s), 802.11n (Wi-Fi 4, up to 600 Mbit/s), 802.11ac (Wi-Fi 5, up to about 6.9 Gbit/s), and 802.11ax (Wi-Fi 6, up to about 9.6 Gbit/s). These are ideal maxima. More efficient spectrum use and multiple spatial streams contribute to the evolution; a maximum rate is not guaranteed to an individual user.

Operator Wi-Fi calling uses Wi-Fi as access to the operator's calling service. It differs from an Internet app call and is not a capability exclusive to Wi-Fi 6. The slide's 9.6 Gbit/s Wi-Fi 6 figure is a theoretical maximum, not normal throughput per device.

LEO and GEO satellite access

Reading basis: NI3 pp. 2-4; slides 01 pp. 42-45

A satellite access system uses radio between Earth terminals and satellites. Distance matters because a signal needs time to propagate. Satellite motion matters because the serving satellite and the available links can change even when the user remains still.

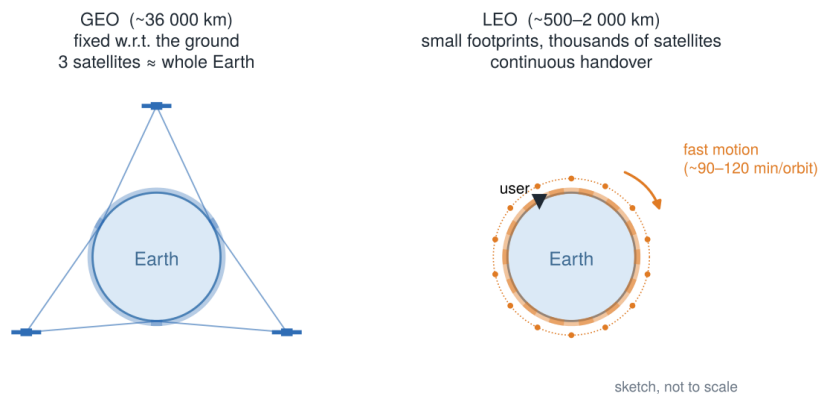


Figure 8. Original NI3 orbit sketch, PDF p. 2. It is not to scale. GEO is distant and appears fixed; LEO satellites are closer, move across the sky and serve smaller footprints.

Property	GEO	LEO
Altitude in course material	About 35,786 km	About 500–2,000 km
Appearance from ground	Fixed over the equator	Moves across the sky
Orbit period	One sidereal day	Roughly 90–120 minutes
Footprint	Large	Smaller
Propagation delay	High	Lower
Continuity problem	Satellite appears stationary	Frequent handovers and constellation coordination

A geostationary satellite has a circular equatorial orbit and matches Earth's rotation. Three appropriately spaced GEO satellites can cover much of the Earth, but do not provide ideal global polar coverage. LEO constellations need many satellites for continuous broad coverage; the exact number depends on design.

For LEO, a stationary user still needs handovers because the satellite moves away. If inter-satellite links are used, routing must account for a changing topology. Not every satellite architecture requires every flow to cross multiple satellites: gateways and system design determine the path.

LEO reduces propagation distance. It does not remove processing, queueing, radio access or terrestrial-network delays. Treat quoted system latency as an end-to-end example, not simply altitude divided by light speed.

Satellite delay and the copper local loop

Reading basis: N13 pp. 3–6; slides 02 pp. 2–10

A worked propagation example

Propagation delay is distance divided by propagation speed. For a rough radio calculation, use $c \approx 300,000$ km/s. One overhead GEO ground-to-satellite hop is $35,786 / 300,000 \approx 0.119$ s, or 119 ms. User → satellite → ground gateway needs two such hops, about 239 ms one way. A request and reply repeat that route, giving about 477 ms before other delays.

At 550 km altitude, an ideal overhead hop takes about 1.83 ms. Two hops take about 3.67 ms one way. Actual slant paths are longer and the complete service contains additional delays. This explains the LEO advantage without treating the ideal number as measured Internet latency.

Why return to telephone copper

The second half of Lesson 3 begins **xDSL**, the family of Digital Subscriber Line technologies. The x identifies the variant: IDSL, HDSL, SDSL, ADSL, RADSL, VDSL and VDSL2. The central idea is to carry digital data over the existing subscriber telephone line.

The **local loop** connects the subscriber premises to the exchange, possibly through passive distribution cabinets. In the historical network, voice on this segment was analog; it was digitised at the exchange. DSL places modulated digital data on the local copper pair, above the low-frequency voice allocation.

A **twisted pair** contains two conductors twisted together. Each subscriber normally has a dedicated pair. Many pairs travel together in a **binder group**. Twisting and differential reception reduce unwanted coupling, but cannot eliminate all interference.

A dedicated copper pair is not a guarantee of dedicated capacity throughout the Internet path. The DSLAM uplink and later network resources are shared. Neighbouring dedicated pairs can also interfere electrically through crosstalk.

Historical mean CO-to-user distance in the slides is about 1.5 km, or 1.2 km in urban areas. These values explain the deployment example; they are not specifications requiring every DSL loop to have that length.

Shannon capacity step by step

Reading basis: NI3 pp. 7–9; full summary PDF p. 10

Shannon's channel-capacity expression relates three quantities: available bandwidth, received signal quality and the ideal maximum information rate. For a band with approximately uniform SNR, the course uses:

$$C = B \log_2(1 + S/N)$$

C: capacity in bit/s. B: bandwidth in Hz. S: received signal power. N: noise power. S/N: a linear ratio, not a value in decibels.

The logarithm is base 2 because information is measured in bits. At a fixed SNR, doubling B doubles the expression. At fixed B, improving SNR increases capacity logarithmically. If noise increases with signal power unchanged, SNR and capacity decrease.

Convert decibels before using the formula

SNR in dB = $10 \log_{10}(S/N)$, so **$S/N = 10^{(SNR_{dB}/10)}$** . Thus 10 dB means a ratio of 10, 20 dB means 100, 30 dB means 1,000 and 40 dB means 10,000. A 0 dB SNR means signal and noise have equal power; it does not mean zero signal.

Worked example with the voice channel

Using the notes' illustrative $B = 3,400$ Hz and $SNR = 30$ dB: convert to $S/N = 1,000$, then compute $C = 3,400 \log_2(1,001) \approx 33,900$ bit/s = 33.9 kbit/s. At 40 dB, $C \approx 45.2$ kbit/s. These are ideal calculations under the stated assumptions, not exact limits for every historical dial-up modem architecture.

Worked example with a wider band

If $B = 1$ MHz and $S/N = 100$, $C = 1,000,000 \log_2(101) \approx 6.66$ Mbit/s. If B becomes 2 MHz with the same SNR, $C \approx 13.32$ Mbit/s. If instead the signal power doubles with B unchanged, S/N becomes 200 and $C \approx 7.65$ Mbit/s. Bandwidth and power affect the expression differently.

Shannon capacity is an ideal limit approached using suitable coding at arbitrarily small error probability. A practical DSL line also has finite constellations, overhead, margins and standards limits. It does not necessarily run at C.

Frequency bandwidth and attenuation

Reading basis: NI3 pp. 8–9 and 16–17; NI4 pp. 7–8

To move beyond voice-band modems, DSL uses a much wider frequency range on the same copper. Do not interpret this as “higher frequency directly makes data faster”. The benefit is **more usable bandwidth**. The cost is that the channel becomes more difficult at those higher frequencies.

A causal chain you should be able to explain

A longer copper loop attenuates the signal more. Its received signal power falls. With comparable noise, the received SNR falls. Each affected tone must carry fewer bits or be disabled. The total attainable bit rate decreases. The frequency allocation need not change merely because the subscriber lives farther from the exchange.

Similarly, high-frequency tones on copper generally suffer more attenuation than low-frequency tones. A short loop may use them productively; a long loop may receive them too weakly. This is why increasing the upper frequency limit works best when copper is also shortened.

Flat and non-flat channels

A **flat channel** has an approximately constant response over the band being considered. A **non-flat**, or frequency-selective, channel affects different frequencies differently. A deep notch may make part of the spectrum unusable. “Flat” does not mean no attenuation: a band can be equally attenuated across all of its frequencies.

The low-frequency voice allocation is narrow enough for simple treatment. A wide ADSL band has much more variable attenuation and noise. Its SNR is therefore not accurately represented by one uniform number across all frequencies. DMT later exploits the quality of each narrow part separately.

Two limits of simplified rules

Doubling bandwidth doubles Shannon’s expression **if SNR is held constant**. With a fixed total transmit power and more integrated noise over the wider band, SNR may change. Also, distance lowers practical rate but there is no universal law saying rate is exactly proportional to $1/\text{distance}$.

Keep these quantities separate: frequency identifies where the signal is, bandwidth identifies how wide the band is, and attenuation/SNR describes how well the signal survives the channel.

ADSL asymmetry and the frequency plan

Reading basis: NI3 pp. 9–10; slides 02 pp. 14–17

ADSL means Asymmetric Digital Subscriber Line. Its downstream rate is generally higher than its upstream rate because the design assigns more resources to downstream. Web browsing and video reception motivated this choice: a small request can produce a large response.

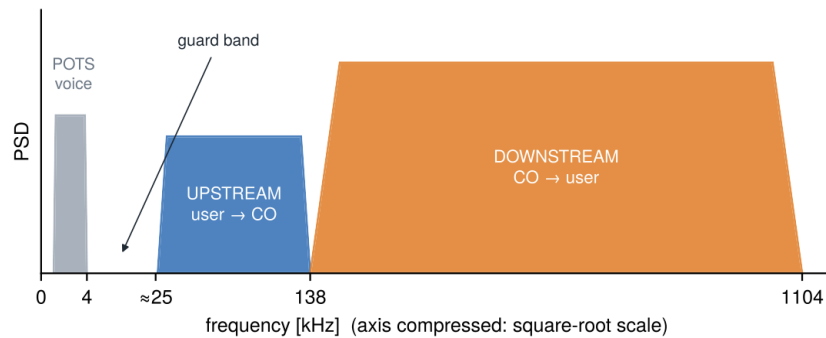


Figure 9. Original frequency-division band plan from NI3, PDF p. 9. Read the direction labels before the numbers. The compressed horizontal scale makes the voice allocation visible.

Allocation	Course band edges	Direction or purpose
POTS	0–4 kHz allocation	Plain Old Telephone Service
Upstream	25.875–138 kHz	User → CO
Downstream	138–1,104 kHz	CO → user

The useful speech band is narrower than the rounded 0–4 kHz allocation. A guard region separates voice from DSL. Filter roll-off and guard intervals keep spectral leakage manageable: practical spectra do not have perfectly sharp rectangular edges.

The downstream frequency-division band is about 966 kHz wide, while the upstream band is about 112 kHz wide. More downstream bandwidth provides more potential capacity, although the actual rate also depends on SNR and the supported configuration.

Why the telephone can coexist with data

The voice signal and DSL occupy different portions of the spectrum on the same pair. Filters separate them at the endpoints. This differs from a dial-up modem that operates within the telephone voice channel and historically uses a call to carry data.

For this conventional ADSL band plan: voice first, upstream next, downstream above it. Asymmetry concerns unequal direction rates; it does not mean communication occurs in only one direction.

The exact figures refer to the ADSL course example. Different annexes, standards and later DSL variants have different plans. The schematic professor slide uses rounded edges; use the more precise table for this guide’s numerical references.

Crosstalk and the NEXT FEXT distinction

Reading basis: NI3 pp. 10–13; slides 02 pp. 18–23

Signals on nearby pairs can couple electromagnetically. The unwanted component received on another pair is **crosstalk**. It depends on coupling, frequency, interferer power and how many active disturbing lines are present. Separate pairs are dedicated physical channels, but they are not perfectly electrically isolated.

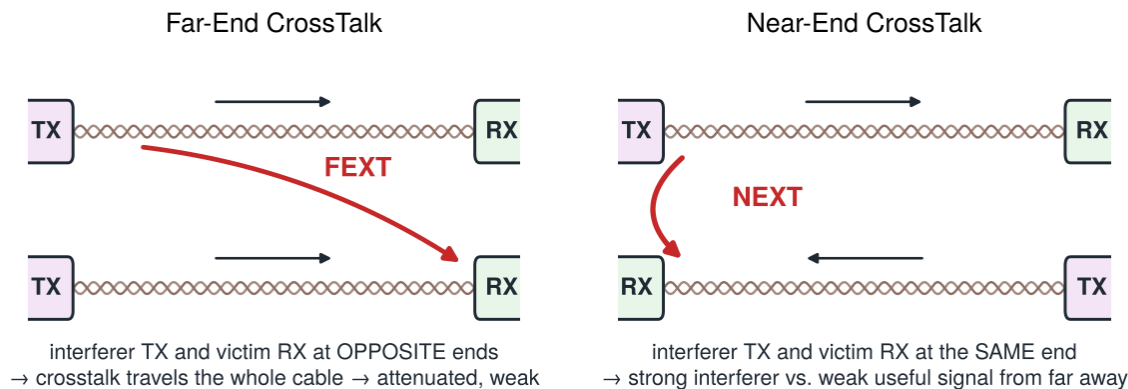


Figure 10. Original NI3 diagram, PDF p. 11. Identify the interfering transmitter and the victim receiver: opposite ends means FEXT; the same end means NEXT.

NEXT is Near-End Crosstalk: the interfering transmitter is near the victim receiver at the same cable end. The interferer can be strong while the wanted signal has already travelled a long attenuating path. **FEXT** is Far-End Crosstalk: the disturbing transmitter is at the opposite end from the victim receiver. The coupling contribution propagates along the line.

Which transmissions conflict

In the course's simplified model, significant direct spectral interference requires simultaneous activity, overlapping frequency resources and coupling between the lines. With ideal separation of upstream and downstream bands, a local downstream transmitter does not inject in-band NEXT into an upstream receiver. Real filtering has finite isolation.

Two upstream lines can create FEXT at the CO. Two downstream lines can also create FEXT at the subscribers, because coupling can occur along a shared cable before the pairs separate. Physical separation of the houses does not eliminate coupling that already happened in the common binder.

Course emphasis and technical limit

NEXT is the major motivation for the ADSL direction-band separation. The lecture treats FEXT as less critical in its introductory example. Do not turn that into "FEXT never matters": VDSL vectoring specifically addresses self-FEXT.

Overlapping bands and cancellation

Reading basis: NI3 pp. 13–15; slides 02 pp. 26–27

Frequency division keeps upstream and downstream separate. An alternative lets downstream extend into part of the upstream spectrum. The downstream then has a wider usable band, but the receiver must cope with interference caused by simultaneous transmission in an overlapping band.

The cancellation principle

If a device knows the signal it transmitted and can estimate the path by which that signal leaks into its own receiver, it can estimate the unwanted component and subtract it from the received waveform. The receiver aims to recover the far-end wanted signal beneath that leakage.

Received signal = wanted signal + leakage + other noise

Subtract an estimate of the known leakage. What remains is the wanted signal, residual cancellation error and other noise. Cancellation does not create a perfectly noise-free channel.

The notes illustrate a CO transmitting downstream on one pair while receiving upstream on a neighbouring pair. Because the CO knows the local transmitted signals, coordinated processing can in principle estimate their coupling and cancel contributions. A customer receiver does not normally know neighbouring customers' transmitted data.

An important terminology distinction

Classical **echo cancellation** removes leakage or echo of a transceiver's own transmitted signal into its receiver on the same duplex link. It can be implemented at either endpoint. Cancelling signals coupled from **other** subscriber pairs is multi-line crosstalk cancellation and requires their signal references and coupling estimates. The notes use the echo-cancellation term loosely for the CO multi-pair example.

For the class comparison, retain the intended result: overlapping direction bands can increase usable downstream bandwidth, but they require more signal processing than ideal frequency separation. Do not infer that a normal single-line echo canceller automatically removes every neighbouring pair's interference.

Why overlapping bands and TDD are different

Overlapping duplex operation permits transmission and reception at the same time, with cancellation. TDD separates the directions in time. Both can reuse frequencies, but their method of avoiding self-interference is different.

More bandwidth is the benefit. Channel estimation, processing, residual interference and implementation complexity are the price. "Only the CO can perform echo cancellation" is too broad; the neighbour-signal example has a specific information constraint.

DMT turns a difficult band into many tones

Reading basis: NI3 pp. 16–19; slides 02 pp. 28–35

A wide copper band can contain good frequencies, poor frequencies and notches. **DMT**, Discrete Multi-Tone, divides that band into many narrow tones and distributes the data across them in parallel. Each sufficiently narrow tone sees an approximately flat channel, making adaptation easier.

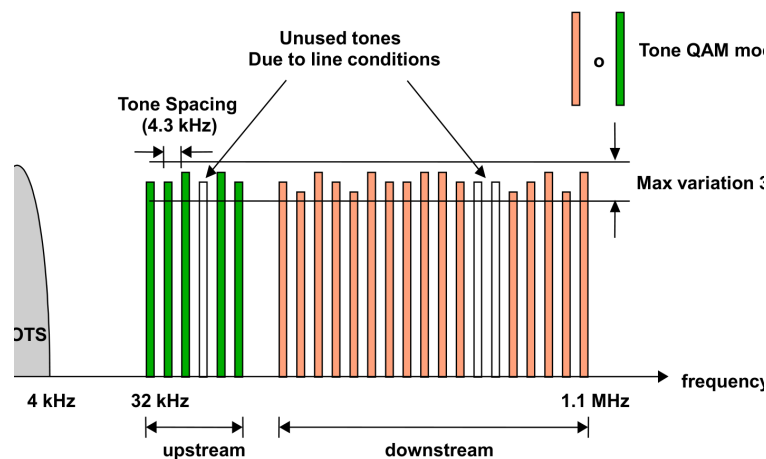


Figure 11. Original professor slide, slides 02 PDF p. 31. Notice equal tone spacing, unequal loading and unused tones. This is a schematic PSD diagram, not a measured SNR trace.

A good tone supports a larger constellation and more bits per symbol. A poor tone carries fewer bits. An unusable tone can be switched off. The receiver reconstructs the original bit stream from the active parallel substreams. Splitting into tones does not physically improve a poor channel; it lets the system use good parts efficiently.

CAP and DMT

CAP, Carrierless Amplitude/Phase modulation, is the single-carrier alternative discussed in the slides. It needs equalisation to cope with a frequency-selective channel. The lecture's "design for the worst frequency" explanation motivates DMT's advantage, but a real single-carrier equaliser is more capable than treating the whole band as one worst point.

DMT adapts loading tone by tone and can avoid narrowband disturbances. Its price is digital signal processing, transform operations, loading information and overhead. It is closely related to the orthogonal multicarrier principle used in OFDM systems.

DMT specifies a multicarrier structure. QAM specifies how symbols carry bits on a tone. They work together; they are not competing modulation options in this architecture.

DMT numbers and rate accounting

Reading basis: NI3 pp. 17–19; slides 02 pp. 29–33

The ADSL example uses a **4.3125 kHz tone spacing** and a grid extending to about **1.104 MHz**: $256 \times 4.3125 \text{ kHz} = 1,104 \text{ kHz}$. The “1.024 MHz” printed on slide 29 is an arithmetic typo. The useful data-symbol rate in the course estimate is approximately **4,000 symbols/s per tone**, which differs from tone spacing because of waveform overhead.

Why 25 plus 249 can exceed 256

The slides quote roughly 25 upstream tones and 249 downstream tones. These counts describe the overlapping-band configuration: the same frequencies can serve both directions with appropriate duplex handling. They are not 274 disjoint tones in a frequency-division plan. Conventional separated upstream/downstream uses different tone allocations. Pilots, reserved tones and standards details further affect actual data loading.

Gross illustrative rate = symbols per second $\times$ sum of bits per symbol over active tones

$R \approx 4,000 \times \sum b_k \text{ bit/s}$. Equal tone width does not imply equal b_k .

The slide’s ideal arithmetic

If every quoted upstream tone carries 15 bits per symbol, $R \approx 25 \times 15 \times 4,000 = 1.50 \text{ Mbit/s}$. For 249 downstream tones, $R \approx 249 \times 15 \times 4,000 = 14.94 \text{ Mbit/s}$. These are idealised loading calculations from the lecture, not guaranteed rates for a standard ADSL service.

For a smaller example, four active tones carrying 2, 4, 6 and 0 bits per symbol give $R \approx 4,000 \times (2 + 4 + 6 + 0) = 48,000 \text{ bit/s}$. The last tone is unused. Adding the rates of active tones gives the total; averaging their bit counts without counting tones would be wrong.

Capacity versus actual bit loading

An idealised per-tone capacity is $C_k = \Delta f \log_2(1 + \text{SNR}_k)$, and the capacities add. Real bit loading must satisfy a target error rate, integer constellation choices, power restrictions and safety margins. Use symbols/s for practical bits-per-symbol accounting and Hz for Shannon bandwidth accounting.

Memorise 4.3125 kHz spacing, about 4,000 useful symbols/s and the idea of unequal bits per tone. Always label the configuration and assumptions when using the 25/249 counts.

Water filling explained as a pool

Reading basis: NI3 pp. 19–22; slides 02 PDF p. 33

Once the band is split into tones, the transmitter must decide how to distribute limited power. **Water filling** is an ideal allocation principle that favours useful tones without spending all available power on a single one.

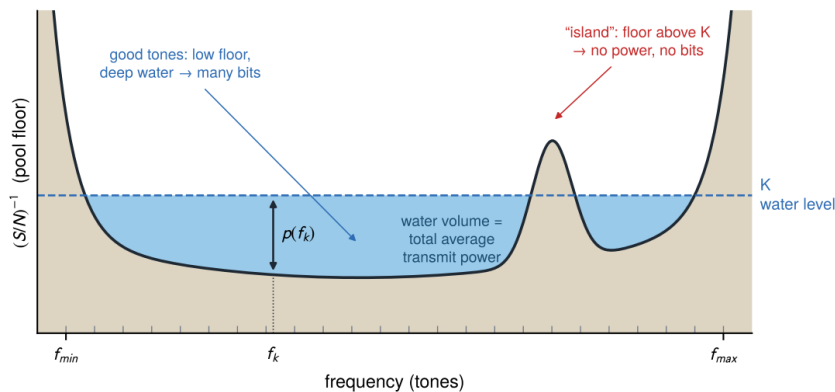


Figure 12. Original NI3 water-filling picture, PDF p. 20. Read the floor as inverse channel quality, the water depth as allocated power and K as the common level.

Picture the tones as adjacent parts of a pool. A low floor represents a good tone: little equivalent noise for each unit of transmitted signal. A high floor represents a poor tone. Pour in a fixed water volume, representing total available power. The surface settles at a common level K .

The depth above a tone's floor is its allocated power. A low floor gets more water. A high floor gets less. If the floor is above K , it becomes an island: no power is assigned, and that tone carries no data.

The course figure's vertical quantity is $(S/N)^{-1}$, not S/N .

The inverse is crucial: high quality means a low floor. Plotting SNR as the floor would reverse the intended intuition.

A precise version of the notation

Let g_k be the received SNR per unit transmit power for tone k . The floor is $1/g_k$ and the power is $p_k = \max(0, K - 1/g_k)$. Choose K so that the allocated powers sum to the total budget. The actual resulting SNR is $p_k g_k$. This avoids treating the final power-dependent SNR as a fixed pool floor.

Power loading and bit loading are related but different: power affects SNR; SNR and the required BER then determine feasible constellation sizes. Practical DSL loading also respects integer bits, spectral masks and margins, so it need not implement unconstrained ideal water filling exactly.

Water filling and QAM worked examples

Reading basis: NI3 PDF p. 21; NI4 pp. 2-4; slides 02 PDF p. 75

Water filling on four tones

Suppose the floors are 1, 2, 4 and 8 in consistent illustrative power units and the available power is 7. Trying all tones gives $4K - 15 = 7$, so $K = 5.5$. Floor 8 lies above that level, so the fourth tone cannot be active. Using the first three gives $3K - 7 = 7$, hence $K = 14/3 \approx 4.67$.

Tone	Floor	Allocated power	Result
1	1	3.67	Best tone gets most
2	2	2.67	Intermediate loading
3	4	0.67	Weak but still active
4	8	0	Island, unused

The powers sum to 7. The point is to understand the allocation, not memorise this particular numerical example. Extra power has diminishing returns through the logarithm, which is why spreading power can be better than putting everything on the best tone.

What a QAM symbol represents

QAM is Quadrature Amplitude Modulation. A constellation contains M possible symbols, represented by points with two coordinates. A symbol selects one point and carries $\log_2 M$ **bits**. The coordinates relate to two quadrature signal components, not two independent subscriber lines.

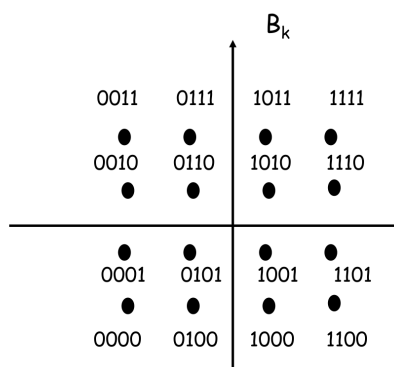


Figure 13. Original 16-QAM constellation from slides 02, PDF p. 75. Sixteen possible points encode four bits per symbol; the binary labels identify symbols.

4-QAM → 2 bits/symbol; 16-QAM → 4; 64-QAM → 6; 256-QAM → 8. “16-QAM” means 16 possible symbols, not 16 bits per symbol.

Bit rate and error rate must be balanced

Reading basis: NI4 pp. 2-4; slides 02 PDF p. 76

To increase rate at a fixed symbol rate, use more possible symbols. But with fixed average signal power, a denser constellation places points closer together. Noise can more easily push a received point across a decision boundary, causing an incorrect symbol or bit.

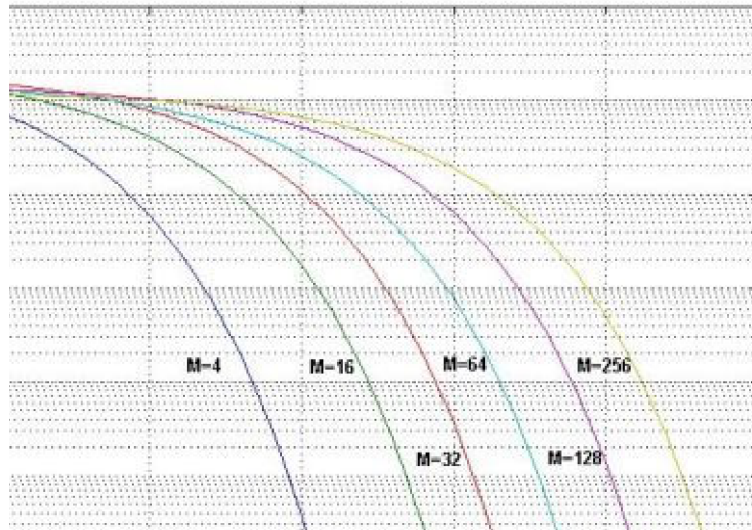


Figure 14. Original BER graph from slides 02, PDF p. 76. The vertical axis is logarithmic BER; the horizontal axis is E_b/N_0 in dB, not ordinary S/N. The curves compare constellation sizes.

How to read the graph

Choose the required BER on the vertical axis and trace a horizontal line across the curves. Each intersection gives the E_b/N_0 needed for that constellation. Alternatively, at a given horizontal value, compare the curves' BERs and choose the densest constellation that still satisfies the target.

At the notes' illustrative $E_b/N_0 = 16$ dB and BER target 10^{-6} , the selected example is 16-QAM: four bits per symbol. A 64-QAM choice would violate the target in that example. At 4,000 symbols/s, four bits per symbol give 16 kbit/s on that tone, before further service overhead.

E_b/N_0 is energy per information bit divided by noise spectral density. It is related to SNR but is not numerically interchangeable without accounting for bit rate and bandwidth. When the graph uses E_b/N_0 , retain that label rather than replacing it with generic "SNR".

Higher constellation order increases bits per symbol, but requires better signal quality for the same BER. A good DMT tone can carry many bits; a poor tone uses a smaller constellation or no data.

ADSL devices from the home to the network

Reading basis: NI4 pp. 4–6; slides 02 pp. 37–46

An ADSL line needs compatible transmitting and receiving equipment at both ends. The **ATU-R** is the ADSL Transceiver Unit at the Remote end, the subscriber premises. The **ATU-C** is the Central-office end. The slides also use “termination unit” terminology. Each end both transmits and receives, using the appropriate direction’s tones.

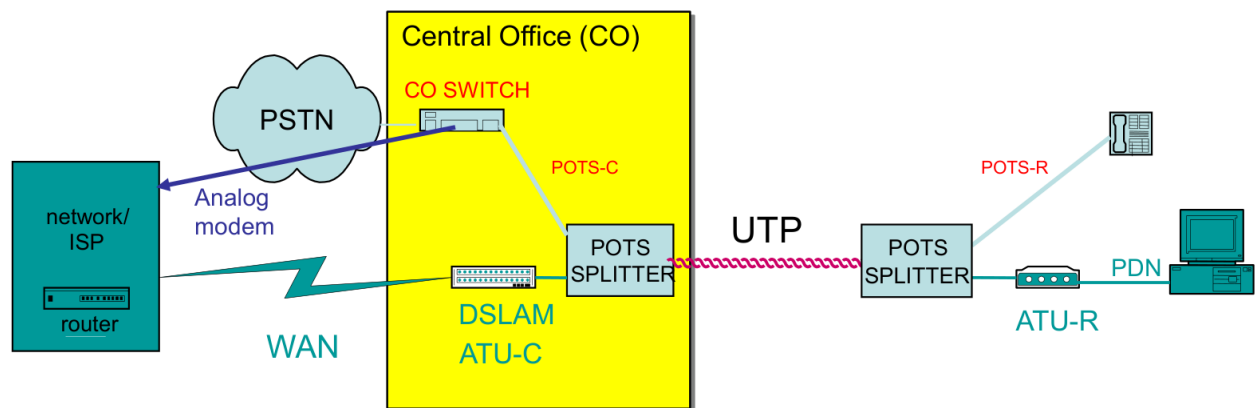


Figure 15. Original ADSL reference model from slides 02, PDF p. 37. Follow the UTP line between POTS splitters, then follow voice to the PSTN and data through ATU-C/DSLAM to the WAN.

A **POTS splitter** separates low-frequency voice from high-frequency DSL. At home, the voice branch reaches the telephone and the DSL branch reaches the transceiver. At the CO, the voice branch reaches telephone switching and the data branch reaches DSL equipment. Filters protect both services from unwanted spectral components and telephone transients.

Why the splitter is not an optical splitter

The POTS splitter separates **frequency bands** on an electrical line. A PON splitter divides **optical power** among branches. They share a word but perform different operations. A POTS splitter does not allocate downstream data to different households.

The familiar home box

A modem performs modulation and demodulation. A household modem-router may combine a DSL transceiver, routing, Ethernet switching and Wi-Fi access. A splitter can be external or integrated; it is not what defines a modem. Ownership also depends on the contract: a modem need not universally belong to the provider.

There must be a DSL line termination for each served pair at the operator side. This means one port/transceiver function per line, not necessarily a separate household-sized box for every subscriber.

DSLAM aggregation and unbundling

Reading basis: NI4 pp. 5–6; slides 01 PDF p. 13; slides 02 pp. 38–44

The **DSLAM** is the DSL Access Multiplexer. It houses many subscriber-side termination interfaces and aggregates their traffic onto uplinks to the provider network. N copper subscriber pairs require N compatible line termination functions, typically integrated into line cards.

Follow a packet in both directions

Upstream: the home ATU-R modulates the transmitted bits. The corresponding ATU-C demodulates them. The DSLAM aggregates traffic from many line ports onto shared uplinks.

Downstream: incoming traffic is directed to the correct subscriber port, whose ATU-C produces that line's DSL waveform.

Aggregation does not mean all users receive the same private packet stream. Port identifiers and higher-layer forwarding keep flows associated with their destinations. A subscriber's physical pair is dedicated, while the aggregation links are shared and provisioned for many users.

Why physical space matters

Line cards, splitters, racks, power supplies and cooling need space. Moving equipment into a street cabinet brings it closer to subscribers but creates the same site-management problem on a smaller scale. An active access node must have power, protection and maintainable physical access.

The unbundling example in the lecture

In the historical **local-loop unbundling** model, another operator uses the incumbent's subscriber copper rather than building a completely separate access plant. To install its own equipment, it may require rented loops, collocation space, power and physical access to the site. The technical lesson is that using a cable also requires somewhere to terminate it.

This describes the model presented in class, not a statement about every operator's current legal entitlement or every wholesale arrangement. Other commercial access models can use shared infrastructure and equipment differently.

The DSLAM aggregates DSL lines. The ATU handles a particular DSL physical link. Unbundling uses another operator's physical access plant; it is not buying a block of free-space radio spectrum.

The ADSL stack and the common IP layer

Reading basis: NI4 pp. 6-7; slides 02 pp. 48-51

A protocol stack explains how a packet becomes a transmittable signal. The historical architecture in the slides uses **IP over PPP over AAL5 over ATM over ADSL**. Read the stack downwards for transmission and upwards for reception.

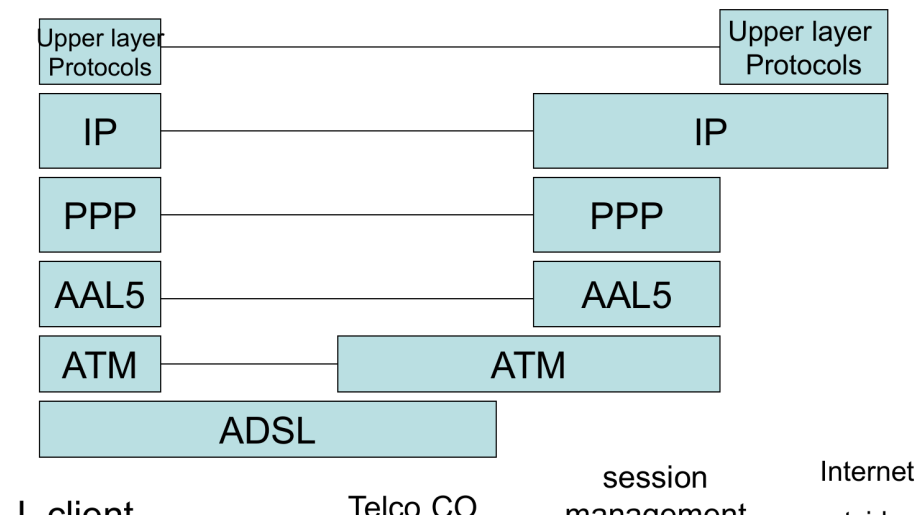


Figure 16. Original stack from slides 02, PDF p. 48. The slide shows physical ATM/ADSL termination at the CO and session management farther into the provider network.

ADSL is the physical transmission layer. **ATM**, Asynchronous Transfer Mode, uses cells as a transport mechanism. **AAL5**, ATM Adaptation Layer 5, adapts larger data units to that cell transport. **PPP** provides point-to-point encapsulation and link/network configuration. **IP** supplies the common network-layer packet format.

On the home LAN, IP packets are instead carried over Ethernet or Wi-Fi. A home router forwards IP traffic between the LAN and provider-facing interface while the lower-layer encapsulation changes. A modem alone can operate as a bridge; routing is an additional function.

Physical termination and session termination differ

The DSL physical line ends at the DSLAM, but the subscriber's PPP session can terminate at a broadband access server farther into the provider network. The notes' "PPP exists only on the copper" wording is therefore too restrictive. Do not assume the DSLAM must terminate PPP or route every subscriber IP packet.

PPPoA means PPP over ATM. PPPoE means PPP over Ethernet. These are encapsulation choices; PPP is not the electrical modulation on copper. The illustrated ATM stack is a historical course architecture, not the universal stack of all DSL deployments.

VDSL shortens copper and widens the band

Reading basis: NI4 pp. 7–10; slides 02 pp. 54–60

ADSL's reuse of long telephone loops has limits. In the lecture's FTTC evolution, fiber reaches the street cabinet and the DSL equipment moves closer to subscribers. Only the final segment remains copper. **VDSL** means Very high speed Digital Subscriber Line; VDSL2 is its later standardised evolution.

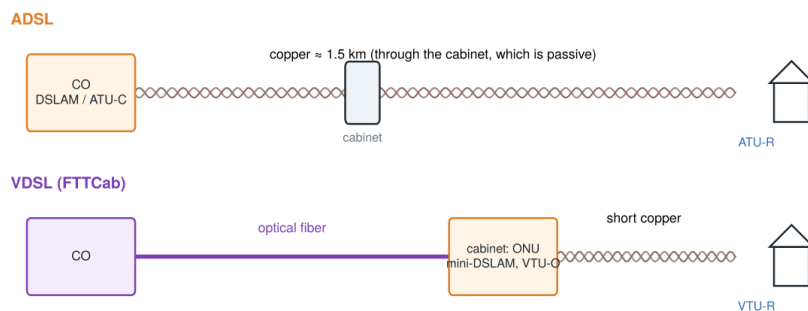


Figure 17. Original ADSL-versus-VDSL layout from NI4, PDF p. 9. In the upper path copper reaches the CO; in the lower path the fiber ends at the cabinet and copper is shorter.

The course names the subscriber-side VDSL unit **VTU-R** and the network/ONU-side unit **VTU-O**. An **ONU** terminates the optical segment in the illustrated hybrid layout. Fiber termination and copper modulation are separate functions, even if housed in the same cabinet.

Two improvements working together

Shorter copper improves received SNR. **A wider spectrum** creates additional potential transmission channels. Shortening the loop allows those extra high-frequency tones to be useful despite their greater attenuation. On a long loop, they may be too poor to contribute much.

Course technology example	Approximate upper frequency
ADSL	1.104 MHz
ADSL2+	2.208 MHz
VDSL example	About 12 MHz
VDSL2 profiles shown	About 17.7 or 30 MHz

These are profile examples from the slides rather than every possible DSL profile. A wider band does not imply the same rate at every distance, and VDSL as a technology is not synonymous with one mandatory FTTC deployment.

The faster final link comes from shorter copper plus more usable bandwidth. Fiber does not change the subscriber's remaining copper into fiber, and higher carrier frequencies alone do not guarantee a larger rate.

FDD and TDD separate the directions differently

Reading basis: NI4 pp. 9–10; slides 02 PDF p. 57

Duplexing specifies how two directions share communication resources. Both directions can be supported using different frequency bands or different time intervals. The VDSL slide compares these ideas; do not interpret every discussed option as the standard mode of every deployed VDSL2 profile.

Property	FDD	TDD
Full name	Frequency Division Duplexing	Time Division Duplexing
Separation	Different frequency bands	Different time intervals
Simultaneous directions	Yes, in different bands	No, for the same time-frequency resource
Use of frequency	Each direction has assigned bands	Directions can reuse the band at different times
Key design concern	Band plan and filters	Timing, guard periods and synchronisation

In a multiband FDD plan, frequency intervals can alternate downstream and upstream. There need not be just one large upstream band and one downstream band. The actual edges depend on the plan and supported profile.

In TDD, a downstream interval is followed by an upstream interval and so on. A direction can use the shared band during its allocated interval. The fraction of time allocated to each direction affects average capacity and asymmetry.

An average-rate example

Suppose the whole band can carry 100 Mbit/s during an active slot. If downstream gets 70% of time and upstream 30%, the ideal average rates are 70 and 30 Mbit/s before guard time and protocol overhead. It would be wrong to say both directions continuously get 100 Mbit/s just because each can use the whole band.

In the overlapping-band ADSL example, simultaneous directions require cancellation. In TDD, their separation in time avoids that simultaneous use on the same link resource. Time synchronisation also matters among neighbouring lines: poorly aligned transmit/receive periods can create strong interference.

FDD separates in frequency; TDD separates in time. “Same band” does not automatically mean interference: activity must also overlap in time and couple into a receiver. The lecture’s historical speculation about why TDD was not chosen for original ADSL is not a general technical law.

Vectoring coordinates several copper lines

Reading basis: NI4 pp. 12–14; slides 02 pp. 62–64

VDSL's high-frequency operation makes crosstalk important. **Vectoring** uses coordinated processing of multiple DSL lines to reduce the interference they cause to one another. The relevant ITU-T G.993.5 mechanism is self-FEXT cancellation: FEXT is not universally harmless.

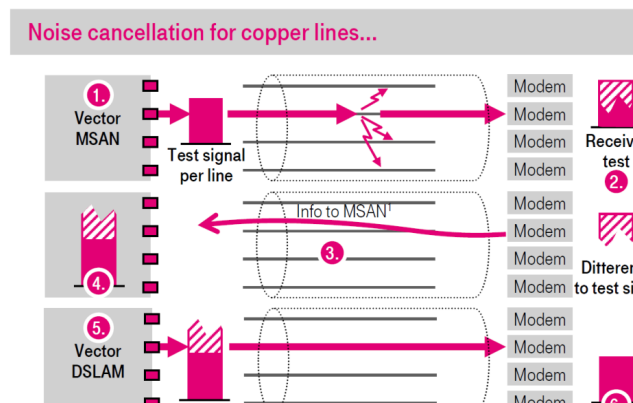


Figure 18. Original vectoring illustration from slides 02, PDF p. 64. Trace the test signal, the receiver's feedback and the adjusted working signal. It is a conceptual illustration.

Known training signals help estimate cross-coupling. Receivers supply information about residual errors. For downstream, the coordinated transmitter can precode the line signals so unwanted coupled contributions cancel at the receivers. For upstream, the coordinated receiving side can apply cancellation after reception.

Why it is a multi-line problem

The processor needs the signals and coupling relationships for the coordinated group. Imagine line 1 receiving $y_1 = x_1 + c x_2$, where $c x_2$ is interference from line 2. In a simplified one-way coupling example, transmitting $x'_1 = x_1 - c x_2$ cancels that contribution. Real systems coordinate a matrix of interactions across many lines and tones.

A test signal arriving with a non-flat response does not by itself prove crosstalk: the direct channel also causes attenuation and distortion. Vectoring specifically estimates and compensates cross-line effects; ordinary single-line equalisation is a different operation.

A fully coordinated group can achieve more cancellation than a partly coordinated one. Uncontrolled disturbing lines can leave residual “alien” crosstalk. Gains depend on line conditions, distance and coordination; vectoring does not guarantee a fixed percentage improvement.

The cost includes processing, synchronisation, training/feedback and compatible equipment. Vectoring helps extend the usefulness of remaining copper; it does not remove the distance-dependent attenuation of that copper.

PPP links and frame fields

Reading basis: NI4 pp. 14–15; slides 02 pp. 66–70

PPP, the Point-to-Point Protocol, transports network-layer packets between two logical peers. It defines encapsulation and mechanisms for configuring the link and supported network-layer protocols. It can be carried over different lower-layer technologies, including Ethernet or ATM.

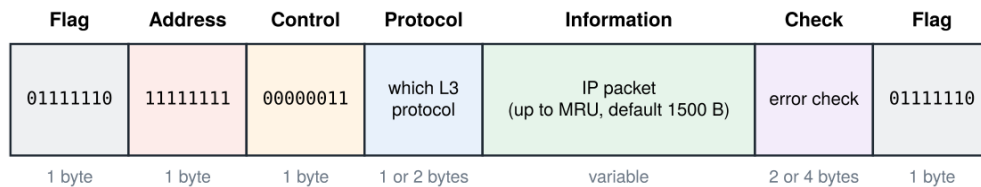


Figure 19. Original classic PPP framing layout from NI4, PDF p. 15. This is HDLC-like framing, not the complete on-wire layout of every PPP encapsulation.

Classic field	Function or course value
Flag	01111110, or 0x7E; frame boundary
Address	11111111, or 0xFF; fixed All-Stations field
Control	00000011, or 0x03; fixed unnumbered-information control
Protocol	Identifies the encapsulated payload protocol
Information	Variable payload; default MRU is 1,500 bytes
FCS or Check	Detects frame corruption; commonly 2 bytes, with a 4-byte option

Point-to-point PPP does not need a destination MAC address to choose between many link peers. Its classic address field is fixed rather than a useful peer identifier. It is therefore wrong to say the pictured frame has no address field; it has a field that does not select a particular destination.

The Information field can carry an IP packet, but PPP control protocols and authentication messages also use PPP encapsulation with their own Protocol values. The Protocol field is therefore broader than “always an IP-version field”.

An FCS detects errors; it does not encrypt the payload or guarantee retransmission. PPP itself is not a reliable-delivery replacement for TCP. PPPoE runs inside Ethernet framing, which still has Ethernet addresses.

PPP establishment and network configuration

Reading basis: NI4 pp. 15–16; slides 02 pp. 68 and 71–73

PPP's lifecycle separates having a physical link, configuring that link, optionally authenticating a peer, and configuring the network layer. Starting a modem is not enough to imply that all of these phases have succeeded.

Phase	What happens	Protocol to remember
Dead	No usable PPP link	Await underlying link
Establishment	Negotiate link options	LCP
Authentication if negotiated	Verify the peer before network access	PAP or CHAP example
Network	Configure network-layer use, then carry packets	NCP family, such as IPCP
Termination	Close the link and release state	LCP

LCP, Link Control Protocol, establishes, configures, tests and terminates the data link. It can negotiate options such as the Maximum Receive Unit, authentication method and compression of certain header fields. Link setup configuration is distinct from the subsequent user-data traffic.

A **Configure-Request** proposes options. A **Configure-Ack** accepts them. A **Configure-Nak** suggests acceptable alternative values. A **Configure-Reject** rejects unsupported options. Both peers participate in reaching a workable link configuration.

NCP is a family of Network Control Protocols for network-layer configuration. **IPCP**, IP Control Protocol, is the IPv4 example. NCPs operate after the link has been established and any required authentication has succeeded.

Authentication is conditional

The course diagram includes an authentication phase because it is relevant to provider access. In general PPP, authentication is optional and is negotiated during establishment. Authentication failure prevents the intended network access rather than silently allowing normal data transfer.

LCP configures the link. NCP configures a network-layer protocol on that link. PAP/CHAP authenticate the peer. IP carries network-layer packets. These are separate jobs even when they all occur in one connection setup.

PAP and CHAP authentication

Reading basis: NI4 pp. 16–17; slides 02 PDF p. 74; RFC 1994

Authentication checks who the peer is. It differs from encryption, which hides data, and from authorisation, which decides what an authenticated peer may do. The lecture compares two PPP authentication methods.

Property	PAP	CHAP
Full name	Password Authentication Protocol	Challenge Handshake Authentication Protocol
Basic exchange	Username and password	Challenge, response, success or failure
Secret sent on the link	Password is sent in clear by PAP itself	Shared secret is not sent
Captured response reuse	Captured credentials can be reused	Fresh challenges impede simple replay
Encrypts user traffic	No	No

With **PAP**, the peer submits credentials and the authenticator accepts or rejects them. A protective underlying tunnel could hide the exchange, but PAP itself does not provide that protection. The course contrast is cleartext credentials versus a challenge-response mechanism.

CHAP step by step

The authenticator sends a challenge. The peer computes a response using a shared secret and that challenge. The authenticator calculates its expected response and compares the values. A match produces success; a mismatch produces failure. A fresh unpredictable challenge prevents simply replaying an old captured response. Challenges can also be repeated during the connection.

The notes write `hash(challenge, password)` as an intuitive abbreviation. In classic RFC 1994 CHAP with MD5, the response is calculated over **Identifier || secret || challenge**, where `||` denotes concatenation. Both sides must have the information needed to calculate the same response.

CHAP improves on sending the password directly, but does not prove that a weak password cannot be guessed offline. A captured challenge-response can be tested against password guesses. Authentication also does not encrypt subsequent PPP/IP traffic.

For this block, learn the exchange, the non-transmission of the secret, and the reason a changing challenge matters. Detailed modern authentication designs are outside the four-lecture scope.

The distinctions that matter most

Do not merge these terms	The precise difference
Bit rate / latency	Amount per second / time to deliver or react
Frequency / bandwidth	Position in spectrum / interval width
Access / core	Subscriber reach / aggregated transport
FTTx / AON or PON	Where fiber reaches / distribution architecture
PON splitter / POTS splitter	Optical branching / electrical frequency separation
Wireless / mobile	Radio medium / continuity while moving
Handover / roaming	Change serving access point / use visited operator
AMF / SMF / UPF	Access and mobility / session control / user packets
NEXT / FEXT	Interfering TX at same end as victim RX / opposite end
Echo / crosstalk cancellation	Own transmitted leakage / interference between lines
DMT / QAM	Many narrow tones / bits encoded in each symbol
Spacing / symbol rate	Hz between tones / symbols per second
ATU / DSLAM	Per-line physical termination / many-line aggregation
FDD / TDD	Separate frequency intervals / separate time intervals
LCP / NCP	Link configuration / network-layer configuration
PAP / CHAP	Transmit credentials / challenge-response with secret

An answer choice can contain a correct term but attach it to the wrong role, direction or quantity. Learn each definition together with what it acts on: a subscriber line, a packet stream, a frequency band, a time slot or a logical session.

When a statement uses “always”, “only”, “never” or “guarantees”, check whether the source described a universal property or only the simplified course example. Exact qualifiers often determine whether a technically similar sentence is correct.

The theory as connected causal chains

Why DSL exists

The telephone network installed copper pairs to many subscribers. Reusing them reduces new deployment costs. The voice allocation is too narrow for modern data demand, so DSL uses higher-frequency bands to obtain more bandwidth. That creates attenuation, frequency selectivity and crosstalk challenges.

Why DMT exists

A wide copper band is not uniformly good. Divide it into narrow nearly flat tones. Assess each tone's quality. Allocate power and bits according to quality and error constraints. Use good tones heavily, bad tones lightly or not at all. Reassemble parallel data at the receiver.

Why VDSL and fiber closer to the user exist

Long copper reduces received SNR, especially at high frequencies. Move the DSL equipment nearer to the user and replace the preceding segment with fiber. Shorter copper makes more high-frequency tones usable. More bandwidth and better SNR increase potential rate. The active cabinet needs space and power.

Why vectoring exists

Higher-frequency DSL increases concern about coupling between lines. A coordinated system estimates cross-line effects. Downstream precoding and upstream cancellation reduce self-FEXT. Uncontrolled lines and residual noise remain possible limits. Vectoring improves copper operation rather than replacing copper.

Why mobility changes architecture

A moving device must share radio resources, remain locatable and keep service when its serving cell changes. Mobility therefore needs signalling and coordination beyond a simple radio link. A LEO system needs handovers even for a fixed user because the serving satellite moves.

Why planes and layers matter

The user plane transports actual service packets; the control plane manages their connectivity. Layer 1 transmits signals, layer 2 frames link traffic, and IP gives a common packet layer across different media. PPP setup, authentication and network configuration prepare the logical link before normal network traffic is carried.

The recurring trade-off is capacity versus deployment cost and transmission quality. Each technique changes a different constraint: fiber changes the medium, shorter copper changes loss, DMT changes spectral use, vectoring changes cross-line interference and PPP changes encapsulation and setup.

Formula and numerical reference

Item	Expression or value	Interpretation
Shannon	$C = B \log_2(1 + S/N)$	B in Hz; linear received SNR
SNR conversion	$SNR_{dB} = 10 \log_{10}(S/N)$	Power ratio to dB
Inverse conversion	$S/N = 10^{(SNR_{dB}/10)}$	Use before Shannon
Propagation delay	$t = d/v$	Use total path distance
Radio speed estimate	$c \approx 300,000 \text{ km/s}$	Ideal propagation calculation
Bits per QAM symbol	$b = \log_2 M$	M is constellation size
DMT gross rate	$R \approx f_s \sum b_k$	f_s in symbols/s
Water filling	$p_k = \max(0, K - 1/g_k)$	g_k is SNR per unit power
POTS allocation	0–4 kHz	Conventional course plan
ADSL upstream	25.875–138 kHz	Frequency-division example
ADSL downstream	138–1,104 kHz	Frequency-division example
Tone spacing	4.3125 kHz	Frequency-grid spacing
DMT useful rate	About 4,000 symbols/s	Distinct from tone spacing
GEO / LEO altitude	35,786 km / about 500–2,000 km	Course orbit comparison
Classic PPP fields	Flag 0x7E; Address 0xFF; Control 0x03	HDLC-like frame example
Default PPP MRU	1,500 bytes	Negotiable receive payload limit

Worked-example values are not measurements of every real line. The 25/249 tone loading and 1.5/14.94 Mbit/s arithmetic assume overlapping direction bands and ideal loading. Service throughput is lower or otherwise limited by the supported standard, overhead and provisioning.

Acronyms and source map

Acronym	Meaning
ADSL / VDSL	Asymmetric / Very high speed Digital Subscriber Line
AAL5 / ATM	ATM Adaptation Layer 5 / Asynchronous Transfer Mode
AON / PON	Active / Passive Optical Network
ATU / VTU	ADSL / VDSL Transceiver Unit
BER / SNR	Bit Error Rate / Signal-to-Noise Ratio
BSC / BTS	Base Station Controller / Base Transceiver Station
CAP / DMT / QAM	Carrierless Amplitude/Phase / Discrete Multi-Tone / Quadrature Amplitude Modulation
CO / DSLAM	Central Office / DSL Access Multiplexer
FDD / TDD	Frequency / Time Division Duplexing
FEXT / NEXT	Far-End / Near-End Crosstalk
FWA / WLL	Fixed Wireless Access / Wireless Local Loop
GEO / LEO	Geostationary Earth Orbit / Low Earth Orbit
LCP / NCP / IPCP	Link Control / Network Control / IP Control Protocol
MSC / RNC	Mobile Switching Centre / Radio Network Controller
OLT / ONU / ODN	Optical Line Terminal / Optical Network Unit / Optical Distribution Network
PAP / CHAP	Password Authentication / Challenge Handshake Authentication Protocol
POTS / PPP	Plain Old Telephone Service / Point-to-Point Protocol
UE / gNB	User Equipment / 5G NR base station

PDF page numbers in captions refer to the page position in the supplied file, including its cover. NI1 covers Lesson 1; NI2 covers Lesson 2; NI3 covers Lesson 3; NI4 covers Lesson 4. The corresponding slide ranges are slides 01 pp. 3–45 and slides 02 pp. 2–76 where relevant to the notes.

Sources and technical clarifications

Primary lecture scope: NI1-dispensa.pdf, NI2-dispensa.pdf, NI3-dispensa.pdf and NI4-dispensa.pdf. Each topic page identifies the specific reading basis.

Professor's material: 01_Infrastructures_Functional_Areas_and_Access_Networks.pdf and 02_Infrastructures_XDSL.pdf. Their original figures are reproduced as selected crops, with captions and explanations. Source images were not redrawn as if they were the originals.

Supplement: Network Infrastructures Theory 24-25 2.5-1-24.pdf. Its early access/DSL material was used selectively and cross-checked; unrelated later topics were not added. Figure 4 comes from its PDF p. 4.

Question-style reference only: Risposte QUIZ II Parte.pdf. The December quiz did not determine subject coverage. No practice questions or answer bank is included.

Corrections that prevent misleading memorisation

The guide distinguishes FEXT significance from the lecture's introductory approximation; neighbour crosstalk cancellation from a single transceiver's own echo cancellation; tone spacing from symbol rate; overlapping tone counts from disjoint FDD allocation; and physical DSL termination from PPP session termination. It also avoids treating all session security as an SMF-only task.

The supplement's wording "decrease noise (SNR)" is corrected: decreasing noise **increases** SNR. Higher frequency is not automatically better signal quality. Dedicated fibers can share ducts or trenches, so one fiber does not inherently require one separate excavation. A passive splitter occupies space and loses optical power even though it needs no electrical supply.

Commercial ownership, deployment examples, spectrum policy comments and operator-company assignments in the notes are not treated as universal technical rules or current market facts. They are unnecessary for mastering this block's engineering principles.

Primary standards used for narrow checks

3GPP technical highlights on 5G security and AUSF

ITU-T G.992.1 ADSL transceivers

ITU-T G.993.5 self-FEXT cancellation for VDSL2

RFC 1661 PPP organisation and control

RFC 1662 PPP in HDLC-like framing

RFC 1994 CHAP challenge and response

RFC 2516 PPP over Ethernet

These checks clarify the supplied lecture material rather than expanding the syllabus. The two-day plan is a study allocation; it cannot predict which questions the professor will select.